

$$\det([T]_B) = \det([T]_{B'}) = \det(T).$$

Theorem. Let V be a finite dimensional vector space over F . Let $T: V \rightarrow V$ be a linear transformation. Then the following statements are equivalent.

- 1) $\lambda \in F$ is an eigenvalue of T .
- 2) The null space of the linear transformation $T - \lambda I: V \rightarrow V$ is non-trivial
- 3) $\det(T - \lambda I) = 0$

Proof: 1) $\Rightarrow$ 2): If λ is an eigenvalue of T , then there exists a non-zero vector $\vec{x} \in V$ such that $T\vec{x} = \lambda\vec{x}$

$$\Leftrightarrow (T - \lambda I)(\vec{x}) = \vec{0}$$

$$\Rightarrow \vec{x} \in \text{Null}(T - \lambda I).$$

$$\Rightarrow \text{Null}(T - \lambda I) \text{ is nontrivial.}$$

2) $\Rightarrow$ 3): Assume $\dim V = n$.

Suppose $\text{Null}(T - \lambda I)$ is nontrivial. Then there exists a nonzero vector $\vec{x} \in \text{Null}(T - \lambda I)$. Extend $\{\vec{x}\}$ to a basis of V : $B = \{\vec{x}, \vec{x}_2, \dots, \vec{x}_n\}$.

$$\text{Then as } (T - \lambda I)(\vec{x}) = \vec{0} \quad [(T - \lambda I)(\vec{x})]_B = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\Rightarrow [T - \lambda I]_B = \begin{pmatrix} 0 & & \\ \vdots & & \\ 0 & & * \end{pmatrix}$$

$$\Rightarrow \det([T - \lambda I]_B) = 0 = \det(T - \lambda I) \quad \text{end of Feb 3}$$

3) $\Rightarrow$ 1). Suppose $\det(T - \lambda I) = 0$.

$\Rightarrow$ there exists an ordered basis $B = \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_n\}$ such that $\det([T - \lambda I]_B) = 0$.

$\Rightarrow [T - \lambda I]_B$ is non-invertible

$$\Rightarrow \text{Null}([T - \lambda I]_B) \neq \{\vec{0}\}$$

$$\Rightarrow \text{there exists a nonzero vector } \vec{a} \in \mathbb{C}^n \text{ such that } [T - \lambda I]_B \vec{a} = \vec{0}.$$

$$\text{take } \vec{x} \in V \text{ such that } [\vec{x}]_B = \vec{a} \neq \vec{0}. \text{ Then } \vec{x} \neq \vec{0}$$

$$\text{then } [T - \lambda I]_B [\vec{x}]_B = \vec{0}$$

$$\Leftrightarrow [(T - \lambda I)\vec{x}]_B = \vec{0}$$

$$\Leftrightarrow (T - \lambda I)\vec{x} = \vec{0}$$

$$\Leftrightarrow T\vec{x} = \lambda\vec{x}.$$

$$\Rightarrow \lambda \text{ is an eigenvalue of } T.$$

Note: Let V be a finite dimensional vector space, and let B be an ordered basis of V . Let $T: V \rightarrow V$ be a linear transformation on V . Then

$$\det(T - \lambda I) = \det([T - \lambda I]_B) = \det([T]_B - \lambda I)$$

Def'n: Let $T: V \rightarrow V$ be a linear transformation with $\dim V = n$. Let B be any ordered basis of V . Then the characteristic polynomial of T is defined to be $\det([T]_B - \lambda I)$.

Remarks: The characteristic polynomial of T is independent from the choice of basis B . As if B' is another basis of V , then there exists

Invertible matrix P such that $[T]_{B'} = P^{-1}[T]_B P$.

$$\text{Then } \det([T]_{B'} - \lambda I) = \det(P^{-1}[T]_B P - \lambda I) = \det(P^{-1}[T]_B P - P^{-1}(\lambda I)P)$$

$$= \det(P^{-1}([T]_B - \lambda I)P) = \det(P^{-1}) \det([T]_B - \lambda I) \det(P)$$

$$= \det(P^{-1}) \det(P) \det([T]_B - \lambda I)$$

$$= \det([T]_B - \lambda I).$$

Proposition: λ_0 is an eigenvalue of T if and only if λ_0 is a root of the characteristic polynomial of T .

Remarks: To find eigenvalues of T , we need to find all the roots of $\det([T]_B - \lambda I) = 0$, i.e. characteristic equation of T .

Remarks: 1). If A is an $n \times n$ matrix, then $\deg(\det(A - \lambda I)) = n$.

2) If $A \in \mathbb{C}^{n \times n}$. Then $f(\lambda) = \det(A - \lambda I)$ can be factored completely into n linear factors, i.e. $\exists \lambda_1, \dots, \lambda_n \in \mathbb{C}$ such that

$$f(\lambda) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda)$$

$\Rightarrow f(\lambda) = 0$ has exactly n roots on $\mathbb{C}$.

Remark: For linear transformations on finite dimensional vector, $T: V \rightarrow V$.

To compute eigenvalues and eigenvectors of T .

① Fix an ordered basis B for V , and compute $[T]_B$.

② Compute eigenvalues and eigenvectors of $[T]_B$

③ Use the basis B and coordinates $\overset{\text{in } \textcircled{2}}{\lambda}$ to recover eigenvalues and eigenvectors of T .

Proposition: Let V be a finite-dimensional vector space over $\mathbb{C}$. Let $T: V \rightarrow V$ be a linear transformation. Let $\lambda_0 \in \mathbb{C}$ be an eigenvalue of T . Then the degree of the factor $(\lambda - \lambda_0)$ in the characteristic polynomial of T is referred as the algebraic multiplicity of λ_0 . The dimension of E_{λ_0} is called the geometric multiplicity of T .

Then the algebraic multiplicity of λ_0 is greater or equal to the geometric multiplicity of λ_0 .

Proof: Suppose the algebraic multiplicity of λ_0 is d . Then the characteristic polynomial of T is of the form $f(\lambda) = (\lambda - \lambda_0)^d g(\lambda)$, where $(\lambda - \lambda_0)^{d+1} \nmid f(\lambda)$. (i.e. the degree of the factor $(\lambda - \lambda_0)$ in $f(\lambda)$ cannot exceed d).
Suppose $\dim E_{\lambda_0} = m$. Then take a basis $\{\vec{u}_1, \dots, \vec{u}_m\}$ of E_{λ_0} .
Then $T(\vec{u}_i) = \lambda_0 \vec{u}_i$ for all $i=1, 2, \dots, m$.

Now extend $\{\vec{u}_1, \dots, \vec{u}_m\}$ to a basis $B' = \{\vec{u}_1, \vec{u}_2, \dots, \vec{u}_m, \vec{v}_{m+1}, \dots, \vec{v}_n\}$ of V .

$$\begin{aligned} \text{Then } [T]_{B'} &= \begin{pmatrix} \boxed{\lambda_0} & & & & & \\ & \ddots & & & & \\ & & \lambda_0 & & & \\ & & & C_{m \times (n-m)} & & \\ & & & 0_{(n-m) \times m} & D_{(n-m) \times (n-m)} & \\ & & & & & \end{pmatrix} \\ \Rightarrow f(\lambda) &= \det([T]_{B'} - \lambda I) = \det \begin{pmatrix} \lambda_0 - \lambda & & & & & \\ & \ddots & & & & \\ & & \lambda_0 - \lambda & & & \\ & & & C & & \\ & & & & (D - \lambda I)_{(n-m) \times (n-m)} & \\ & & & & & \end{pmatrix} \\ &= \det \begin{pmatrix} \lambda_0 - \lambda & & & & \\ & \ddots & & & \\ & & \lambda_0 - \lambda & & \\ & & & C & \\ & & & & (D - \lambda I)_{(n-m) \times (n-m)} \end{pmatrix} \\ &= (\lambda_0 - \lambda)^m h(\lambda). \end{aligned}$$

$$\Rightarrow (\lambda_0 - \lambda)^m \mid (\lambda_0 - \lambda)^d \Rightarrow m \leq d.$$

Note:

$$\begin{aligned} \det \begin{pmatrix} \lambda - \lambda_0 & & & & & \\ & \ddots & & & & \\ & & \lambda - \lambda_0 & & & \\ & & & C & & \\ & & & & D - \lambda I & \\ & & & & & \end{pmatrix}_{n \times n} &\stackrel{\text{expand along first column}}{=} (\lambda - \lambda_0) \cdot \det \begin{pmatrix} \lambda - \lambda_0 & & & & & \\ & \ddots & & & & \\ & & \lambda - \lambda_0 & & & \\ & & & C & & \\ & & & & D - \lambda I & \\ & & & & & \end{pmatrix}_{(n-1) \times (n-1)} \\ &\stackrel{\text{expand along first column}}{=} (\lambda - \lambda_0)^2 \det \begin{pmatrix} \lambda - \lambda_0 & & & & & \\ & \ddots & & & & \\ & & \lambda - \lambda_0 & & & \\ & & & C & & \\ & & & & D - \lambda I & \\ & & & & & \end{pmatrix}_{(n-2) \times (n-2)} \\ &\vdots \\ &= \dots = (\lambda - \lambda_0)^m \det(D - \lambda I) \end{aligned}$$

Proposition: Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of the linear transformation T .

Let $\vec{x}_1, \vec{x}_2, \dots, \vec{x}_k$ be eigenvectors corresponding to $\lambda_1, \lambda_2, \dots, \lambda_k$ respectively.

Then $\{\vec{x}_1, \dots, \vec{x}_k\}$ is linearly independent.

Proof: Let $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_i\} \subseteq \{\vec{x}_1, \dots, \vec{x}_k\}$.

Proof by induction on i .

Base case: when $i=1$ $\{\vec{x}_1\} \neq \{\vec{0}\}$, thus $\{\vec{x}_1\}$ is linearly independent.

Inductive steps. Assume $\{\vec{x}_1, \dots, \vec{x}_i\}$ is linearly independent

$$\text{Suppose } a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_{i+1} \vec{x}_{i+1} = \vec{0} \quad (*)$$

$$\text{Then } T(a_1 \vec{x}_1 + a_2 \vec{x}_2 + \dots + a_{i+1} \vec{x}_{i+1}) = \vec{0}$$

$$\Leftrightarrow \lambda_1 a_1 \vec{x}_1 + \lambda_2 a_2 \vec{x}_2 + \dots + \lambda_{i+1} a_{i+1} \vec{x}_{i+1} = \vec{0} \quad ①$$

$$\text{multiply } \lambda_{i+1} \text{ to } (*): a_1 \lambda_{i+1} \vec{x}_1 + a_2 \lambda_{i+1} \vec{x}_2 + \dots + a_i \lambda_{i+1} \vec{x}_i + a_{i+1} \lambda_{i+1} \vec{x}_{i+1} = \vec{0} \quad ②$$

$$② - ①: a_1 (\lambda_{i+1} - \lambda_1) \vec{x}_1 + a_2 (\lambda_{i+1} - \lambda_2) \vec{x}_2 + \dots + a_i (\lambda_{i+1} - \lambda_i) \vec{x}_i = \vec{0}$$

By inductive assumption that $\{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_i\}$ is linearly independent

$$\Rightarrow a_1 (\lambda_{i+1} - \lambda_1) = a_2 (\lambda_{i+1} - \lambda_2) = \dots = a_i (\lambda_{i+1} - \lambda_i) = 0$$

As $\lambda_1, \dots, \lambda_{i+1}$ are distinct eigenvalues: $\lambda_{i+1} - \lambda_1 \neq 0, \lambda_{i+1} - \lambda_2 \neq 0, \dots, \lambda_{i+1} - \lambda_i \neq 0$

$$\Rightarrow a_1 = a_2 = \dots = a_i = 0$$

$$\text{plug } a_1 = a_2 = \dots = a_i = 0 \text{ into } (*): a_{i+1} \vec{x}_{i+1} = \vec{0}$$

$$\text{Since } \vec{x}_{i+1} \neq \vec{0}, a_{i+1} = 0$$

$$\Rightarrow \{\vec{x}_1, \vec{x}_2, \dots, \vec{x}_{i+1}\} \text{ is linearly independent.} \quad // \text{ end of Feb 5}$$

Corollary. Let $\lambda_1, \lambda_2, \dots, \lambda_k$ be distinct eigenvalues of the linear transformation T .

Let $\vec{x}_1 \in E_{\lambda_1}, \vec{x}_2 \in E_{\lambda_2}, \dots, \vec{x}_k \in E_{\lambda_k}$. If $\vec{x}_1 + \vec{x}_2 + \dots + \vec{x}_k = \vec{0}$,

then $\vec{x}_1 = \vec{x}_2 = \dots = \vec{x}_k = \vec{0}$.